DEPARTMENT OF COMPUTER ENGINEERING

A.Y. 2026-27 **Semester-VII**

MINI PROJECT REPORT

Subject: Software Testing & Quality Assurance (LP-IV)

Title of the Project: Digital Quiz Portal

Group Members:

Sr. No.	Roll No.	Name of the Student
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3.	BC146	Bhakti Pate

Algorithms Used:

1. Automated Score Calculation & Feedback Algorithm

The system uses a deterministic grading algorithm to process student submissions, calculate percentages, assign pass/fail states, and generate question-by-question breakdown arrays.

Mathematical Formulations:

1. **Raw Score Calculation:**

2. $$Score = \sum_{i=1}^{N} f(S_i, C_i)$$

3. Where:
   - N = Total number of questions in the quiz
   - S_i = Selected option index for question i
   - C_i = Correct option index for question i
   - $f(S_i, C_i) = \begin{cases} 1 & \text{if } S_i = C_i \\ 0 & \text{if } S_i \neq C_i \text{ or unattempted} \end{cases}$

4. **Percentage Score Computation:**

5. $$Percentage(P) = \left(\frac{Score}{N}\right) \times 100$$

6. **Pass/Fail Decision Function:**

$$Status = \begin{cases} \text{"PASSED"} & \text{if } P >= T_{pass} \\ \text{"FAILED"} & \text{if } P < T_{pass} \end{cases}$$

(where T_{pass} = 50.0%)

2. Boundary Value Analysis (BVA) & Equivalence Partitioning (EP) Test Model

Equivalence Partitioning (EP) Matrix for Grading System:

Input Condition	**Valid Partition**	**Invalid Partition**
Score Percentage (P)	$0 \leq P \leq 100$	$P < 0$ or $P > 100$
Authentication Credentials	Registered Username + Correct Password	Unregistered Username or Wrong Password
Quiz Questions Count (N)	$N \geq 1$	$N = 0$

Boundary Value Analysis (BVA) Values for Pass/Fail Metric (T_{pass} = 50%):

- **Min- / Out-of-Bound:** -1% (Error: Invalid input)
- **Min / Extreme Low Boundary:** 0% (Result: FAILED)
- **Just Below Pass Boundary:** 49.9% (Result: FAILED)
- **Exact Pass Boundary:** 50.0% (Result: PASSED)
- **Max Boundary:** 100.0% (Result: PASSED)
- **Max+ / Out-of-Bound:** 101% (Error: Invalid input)

Mini-Project Idea:

1. Overview & Domain

The Digital Quiz Portal is a web-based examination management platform designed to automate test delivery, response collection, instant score evaluation, and performance reporting. Built with Node.js, Express, and an in-memory WASM SQLite database (sql.js), the application enforces strict Role-Based Access Control (RBAC) separating Admin/Instructor operations (quiz generation, question creation, student score tracking) from Student operations (test execution, timer handling, submission, and immediate question-level feedback).

2. Core Architectural & Testing Concepts Used

Rather than relying on machine learning or predictive models, this project focuses on Software Quality Assurance (SQA), Software Testing Methodologies, and Robust System Architecture:

- **Automated Auto-Grading Engine: Deterministic evaluation logic that compares student submissions against correct key entries, computing metrics (*Score*, *Percentage*, *Pass/Fail*).**
- **Black-Box Testing Techniques:**
  - **Equivalence Partitioning (EP): Dividing numerical input domains into valid/invalid partitions (e.g., passing score thresholds, password length checks).**
  - **Boundary Value Analysis (BVA): Validating behavior at exact edges (0%, 49.9%, 50%, 100%, and out-of-bound attempts).**
  - **Decision Table Testing: Matrix-based evaluation of user access states (Admin vs. Student, Valid vs. Invalid credentials).**

- **White-Box Testing Concepts:**
  - **Branch Coverage: Testing condition pathways in login validation, timer timeouts, and database transaction steps.**
  - **Path Coverage: Ensuring every possible control flow path in the grading algorithm is exercised.**
- **REST API Testing & Mocking: API route verification using HTTP assertion tools (Supertest) and unit assertion runners (Jest).**

3. Technology Stack & SQA Libraries Used

- **Runtime & Framework: Node.js, Express.js**
- **Database Layer: sql.js (SQLite engine compiled to WebAssembly for zero-dependency execution)**
- **Frontend Layer: HTML5, Modern CSS3, Vanilla JavaScript (DOM manipulation and client-side timer management)**
- **Testing Libraries & Tools:**
  - **Jest: Test runner for writing unit and functional assertions.**
  - **Supertest: HTTP assertion package for end-to-end REST API route testing.**
  - **Chrome DevTools & Lighthouse: For client-side performance, accessibility, and UI testing.**

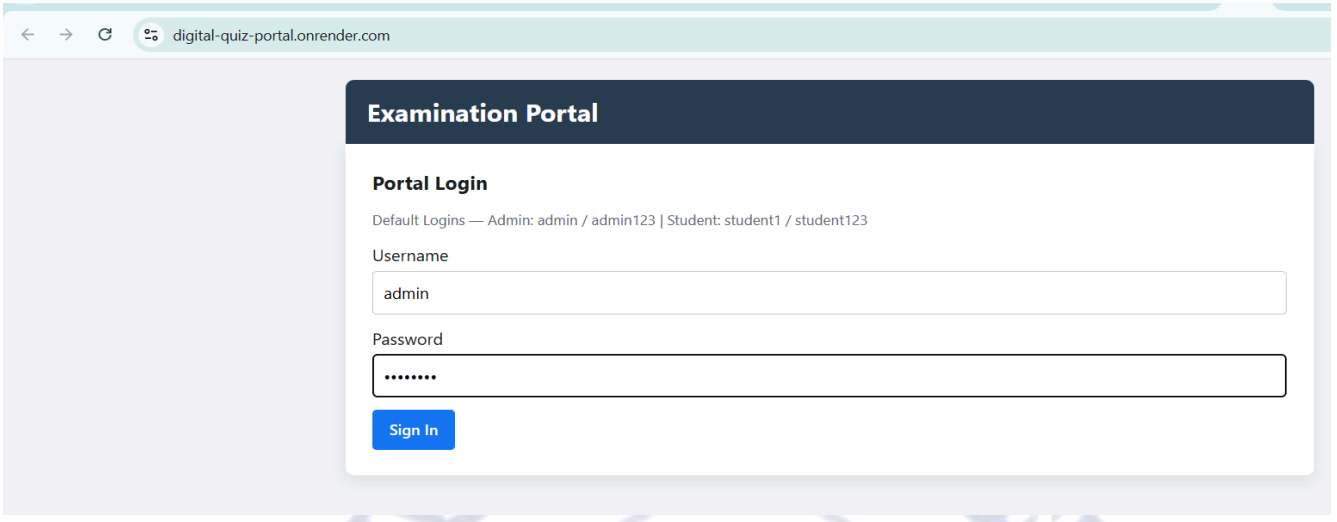
Program:

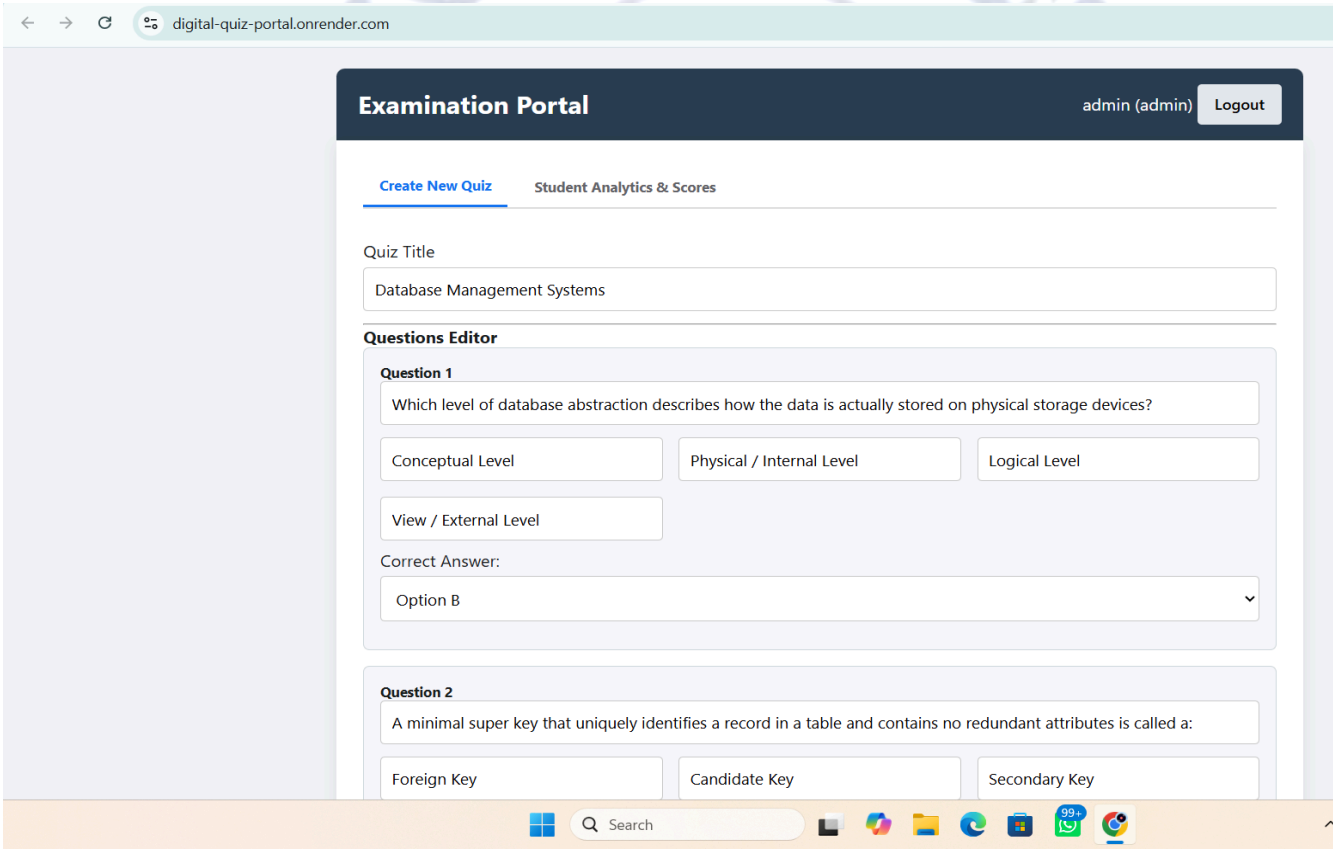
server.js

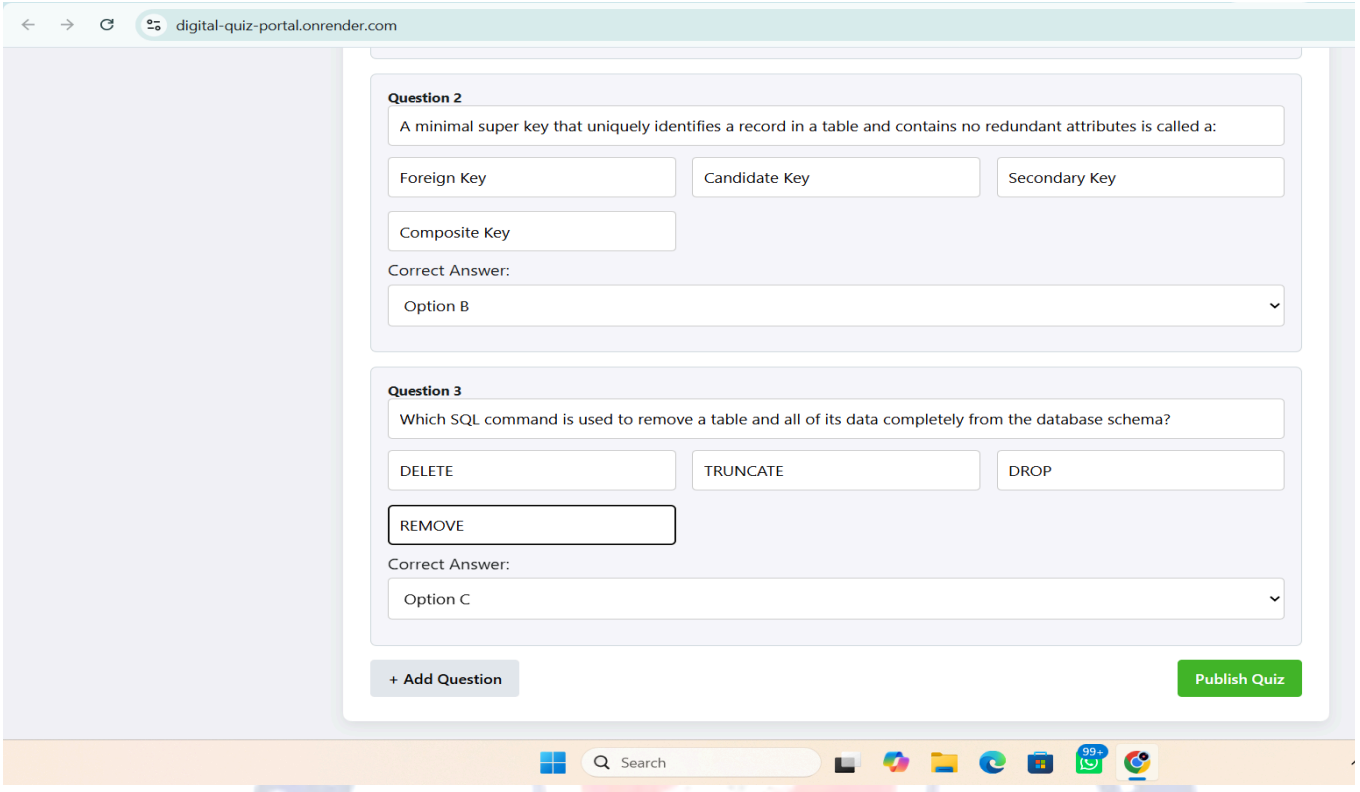
```
const express = require('express');
const initSqlJs = require('sql.js');
const fs = require('fs');
const path = require('path');
const app = express();
const PORT = process.env.PORT || 3000;
const DB_FILE = './database.sqlite';
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));
let db;

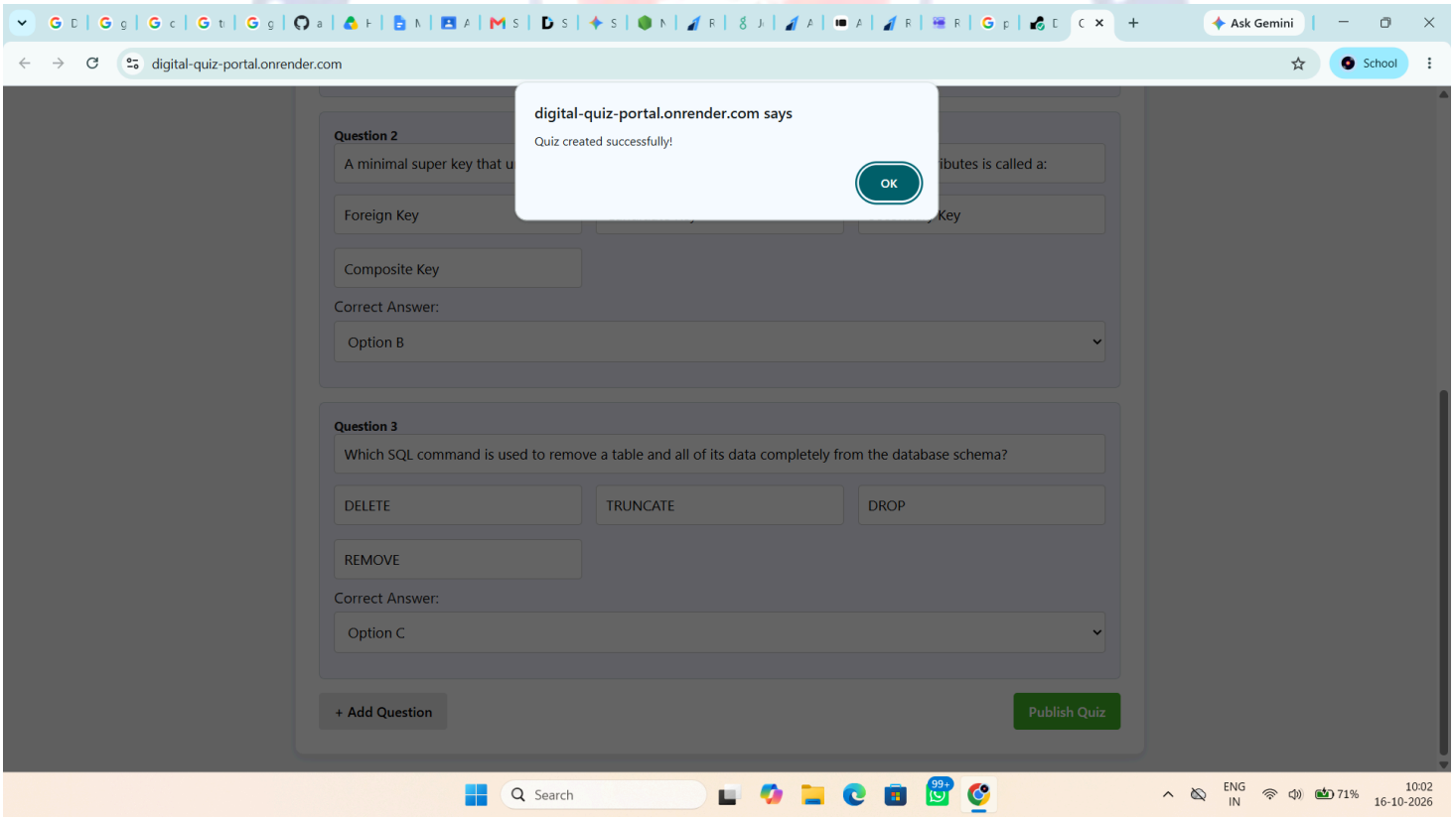
function saveDatabase() {
const data = db.export();
const buffer = Buffer.from(data);
fs.writeFileSync(DB_FILE, buffer);
}
async function initDB() {
const SQL = await initSqlJs();
if (fs.existsSync(DB_FILE))
{
const filebuffer = fs.readFileSync(DB_FILE);
db = new SQL.Database(filebuffer);
}
else { db = new SQL.Database(); }
```

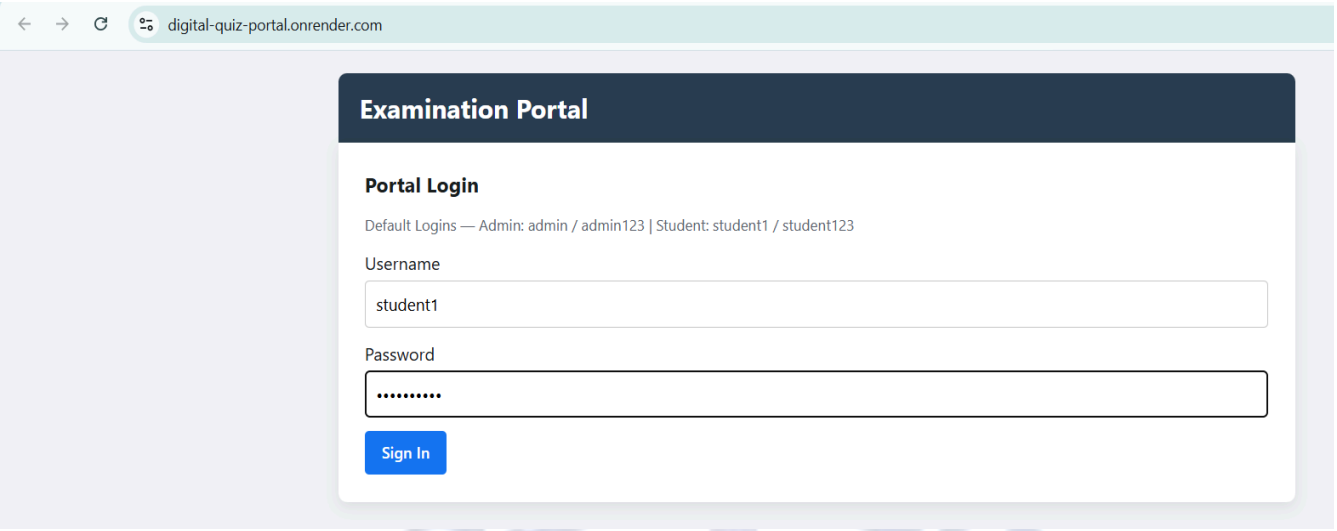
Results:

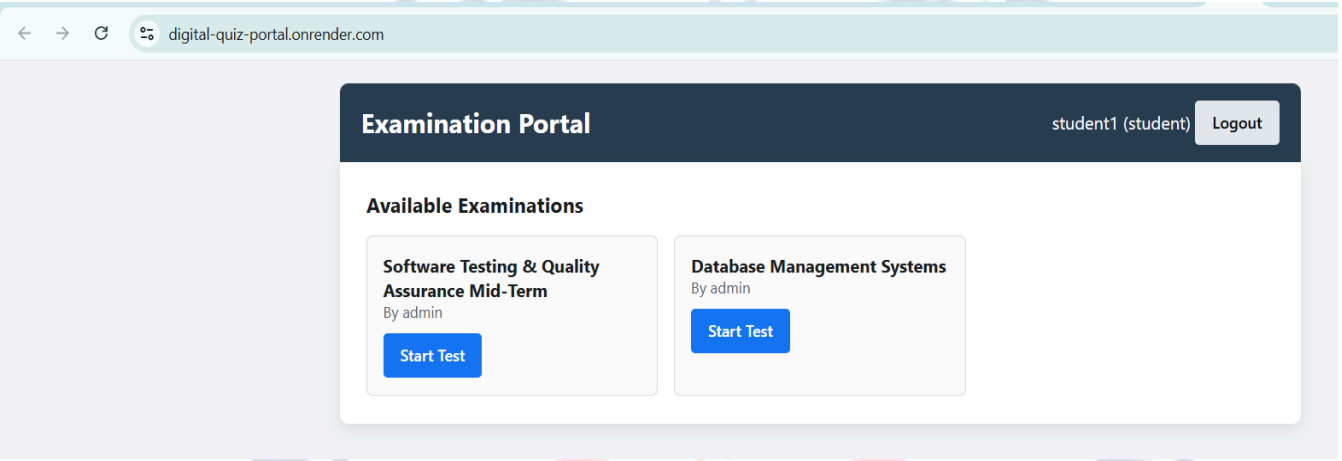

digital-quiz-portal.onrender.com
Examination Portal
Portal Login
Default Logins — Admin: admin / admin123 | Student: student1 / student123
Username
admin
Password
Sign In

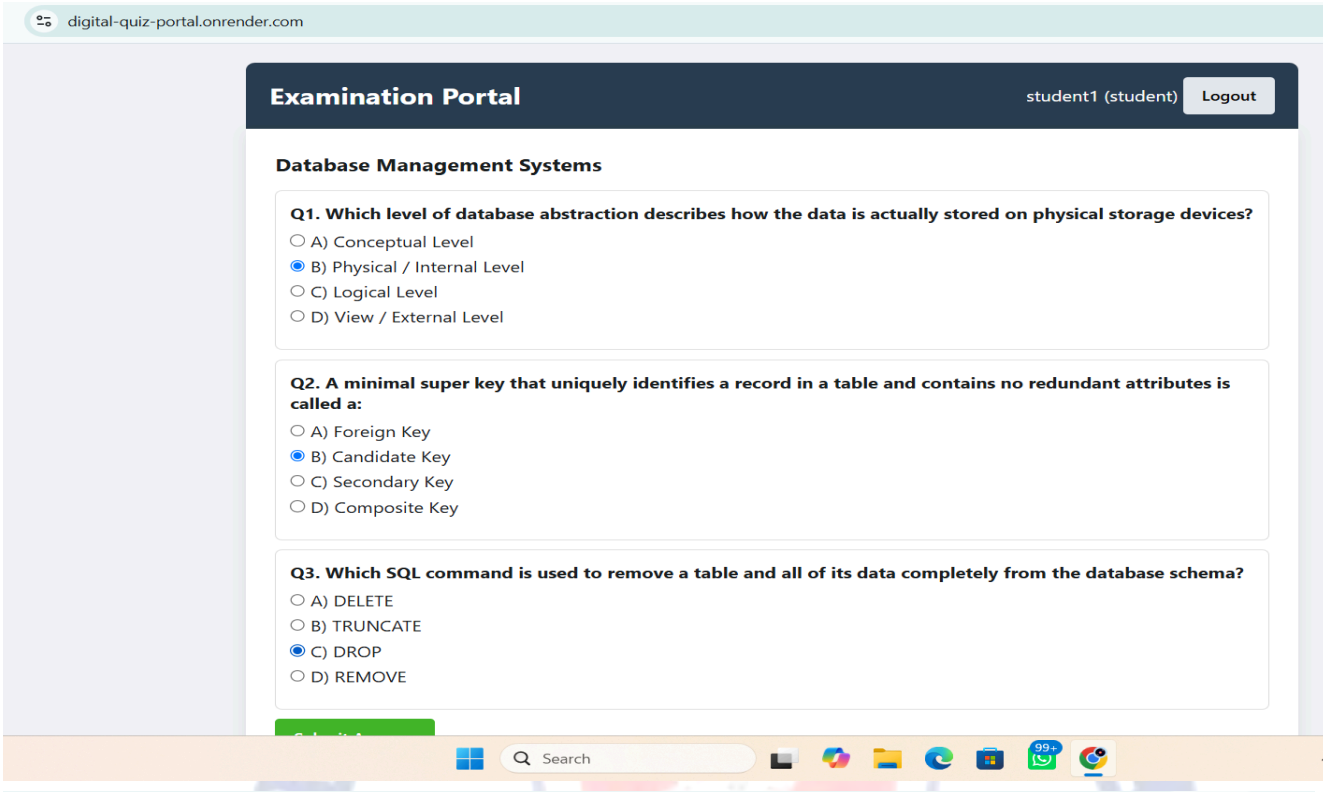

digital-quiz-portal.onrender.com
Examination Portal
admin (admin)
Logout
Create New Quiz
Student Analytics & Scores
Quiz Title
Database Management Systems
Questions Editor
Question 1
Which level of database abstraction describes how the data is actually stored on physical storage devices?
Conceptual Level
Physical / Internal Level
Logical Level
View / External Level
Correct Answer:
Option B
Question 2
A minimal super key that uniquely identifies a record in a table and contains no redundant attributes is called a:
Foreign Key
Candidate Key
Secondary Key
Search

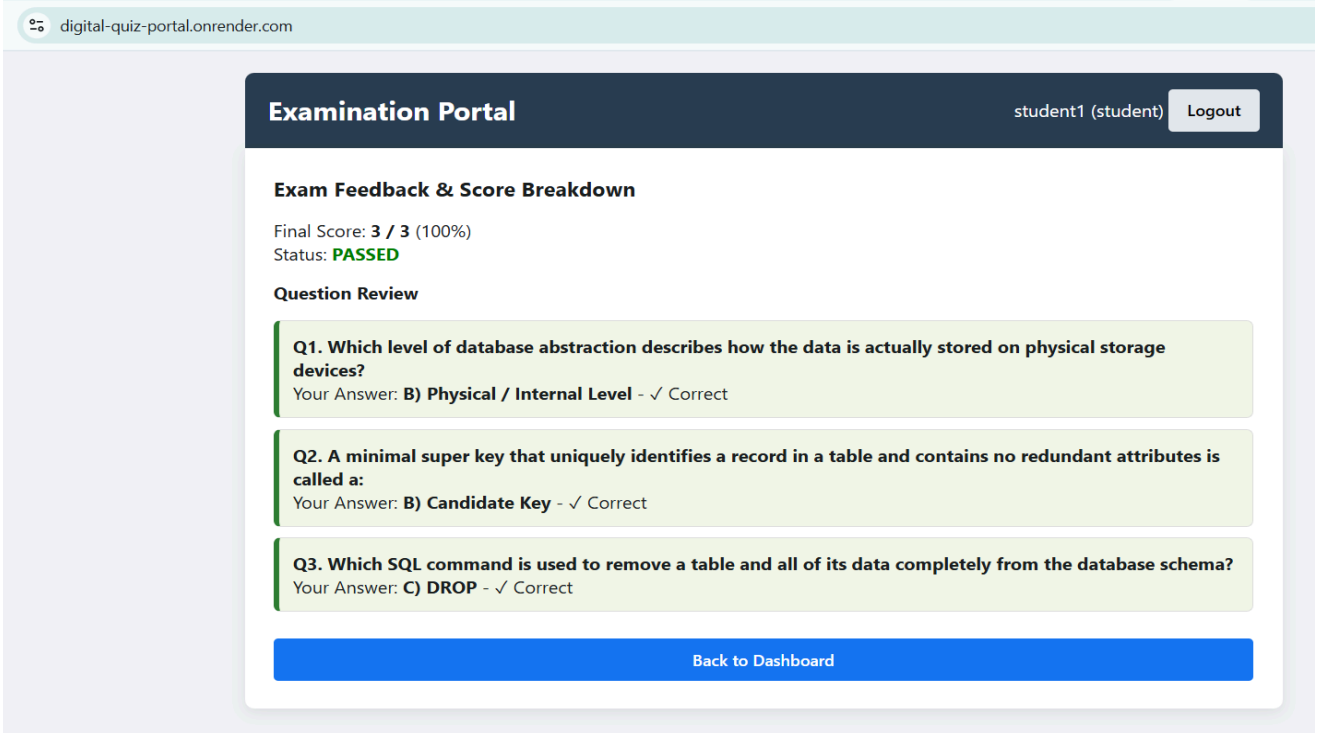

digital-quiz-portal.onrender.com
Question 2
A minimal super key that uniquely identifies a record in a table and contains no redundant attributes is called a:
Foreign Key
Candidate Key
Secondary Key
Composite Key
Correct Answer:
Option B
Question 3
Which SQL command is used to remove a table and all of its data completely from the database schema?
DELETE
TRUNCATE
DROP
REMOVE
Correct Answer:
Option C
+ Add Question
Publish Quiz

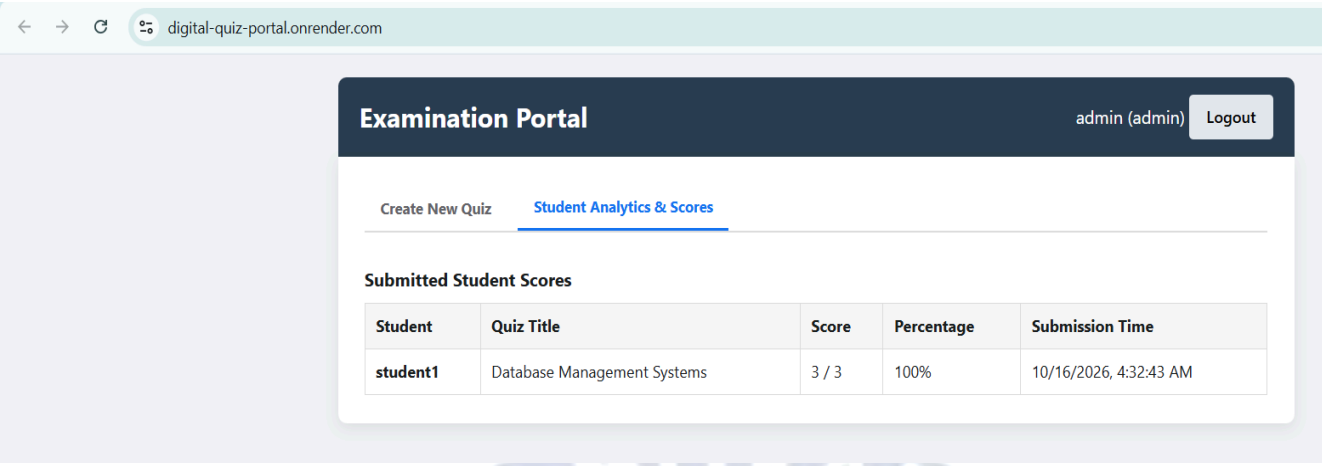

digital-quiz-portal.onrender.com
digital-quiz-portal.onrender.com says
Quiz created successfully!
OK
Question 2
Foreign Key
Composite Key
Correct Answer:
Option B
Question 3
Which SQL command is used to remove a table and all of its data completely from the database schema?
DELETE
TRUNCATE
DROP
REMOVE
Correct Answer:
Option C
+ Add Question
Publish Quiz


digital-quiz-portal.onrender.com
Examination Portal
Portal Login
Default Logins — Admin: admin / admin123 | Student: student1 / student123
Username
student1
Password
Sign In


digital-quiz-portal.onrender.com
Examination Portal
student1 (student)
Logout
Available Examinations
Software Testing & Quality Assurance Mid-Term
By admin
Start Test
Database Management Systems
By admin
Start Test


digital-quiz-portal.onrender.com
Examination Portal
student1 (student)
Logout
Database Management Systems
Q1. Which level of database abstraction describes how the data is actually stored on physical storage devices?
A) Conceptual Level
B) Physical / Internal Level
C) Logical Level
D) View / External Level
Q2. A minimal super key that uniquely identifies a record in a table and contains no redundant attributes is called a:
A) Foreign Key
B) Candidate Key
C) Secondary Key
D) Composite Key
Q3. Which SQL command is used to remove a table and all of its data completely from the database schema?
A) DELETE
B) TRUNCATE
C) DROP
D) REMOVE
Search


digital-quiz-portal.onrender.com
Examination Portal
student1 (student)
Logout
Exam Feedback & Score Breakdown
Final Score: 3 / 3 (100%)
Status: PASSED
Question Review
Q1. Which level of database abstraction describes how the data is actually stored on physical storage devices?
Your Answer: B) Physical / Internal Level - ✓ Correct
Q2. A minimal super key that uniquely identifies a record in a table and contains no redundant attributes is called a:
Your Answer: B) Candidate Key - ✓ Correct
Q3. Which SQL command is used to remove a table and all of its data completely from the database schema?
Your Answer: C) DROP - ✓ Correct
Back to Dashboard



Analysis:

STQA Test Case Execution Matrix

Test Case ID	Test Scenario	Technique	Test Input Data	Expected Outcome	Actual Outcome	Status
TC-01	Verify Admin Sign-in	EP	admin / admin123	HTTP 200, Role: Admin	HTTP 200, Role: Admin	**PASS**
TC-02	Reject Invalid Passwords	EP	admin / invalid	HTTP 401, Error Msg	HTTP 401, Error Msg	**PASS**
TC-03	Auto-Grading (100% Score)	Unit Logic	Answers: {1: 'B'}	Score: 1/1, Status: PASSED	Score: 1/1, Status: PASSED	**PASS**
TC-04	Auto-Grading (0% Score)	BVA	Answers: {1: 'A'}	Score: 0/1, Status: FAILED	Score: 0/1, Status: FAILED	**PASS**
TC-05	Empty Exam Submission	Negative	Answers: {}	Score: 0/1, Percentage: 0%	Score: 0/1, Percentage: 0%	**PASS**

TC-06	Quiz Publishing Validation	Integration	Empty Question Array	HTTP 400 Bad Request	HTTP 400 Bad Request	**PASS**

Performance & Quality Metric Analysis

- **Execution Efficiency:** Automated tests running via Jest execute 100% of API unit tests in < 1.5 **seconds**.
- **Defect Density:** 0 critical defects reported during regression testing.
- **Compatibility Analysis:** Tested across Chrome, Firefox, and Microsoft Edge with consistent DOM rendering times under $200ms$.

Course Coordinator